B4.2 Ecological niches

NAME	DATE	TOTAL
		_____ / 48

About 50 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

10 MARKS

1 **DEFINE** ecological niche. [B4.2.1] [1]

2 **STATE** the type of organism that cannot tolerate oxygen. [B4.2.2] [1]

3 **NAME** three groups of organisms that photosynthesize. [B4.2.3] [1]

4 **LIST** the stages of holozoic nutrition, in order. [B4.2.4] [2]

5 **NAME** one mixotrophic protist. [B4.2.5] [1]

6 **STATE** two ways in which a plant can resist herbivores. [B4.2.9] [2]

7 **STATE** two possible outcomes when two species compete for one niche. [B4.2.13] [2]

Section B: Understanding

17 MARKS

8 **DISTINGUISH** between a habitat and a niche. [B4.2.1] [2]

9 **DESCRIBE** saprotrophic nutrition, and explain its importance to an ecosystem. [B4.2.6] [3]

10 **OUTLINE** the diversity of nutrition in archaea. [B4.2.7] [3]

11 **EXPLAIN** how an aphid obtains its food. [B4.2.9] [3]

12 **COMPARE AND CONTRAST** a liana and an epiphyte. [B4.2.11] [3]

LIANA	EPIPHYTE

13 **SUGGEST** one chemical, one physical and one behavioural adaptation in prey animals. [B4.2.10] [3]

Section C: Working with data

15 MARKS

Data in question 14 are invented for practice, using the pattern in Gause's experiments. Data in question 15 are approximate.

14 [B4.2.12, B4.2.13]

Two species of Paramecium were grown separately, and then together in one flask.

DAY	0	4	8	12	16
SPECIES A, ALONE / CELLS PER CM^3	5	60	180	200	200
SPECIES B, ALONE / CELLS PER CM^3	5	40	110	130	130
SPECIES A, TOGETHER	5	50	140	185	195
SPECIES B, TOGETHER	5	30	40	15	2

a **DESCRIBE** the changes in numbers of species B when grown together with A. [2]

b **DEDUCE** what these data show about the niches of the two species. [2]

c **STATE** the principle shown by these data. [1]

d **SUGGEST** what might have happened if the flask had two separate regions with different food. [2]

15 [B4.2.8]

FEATURE	HOMO SAPIENS	PARANTHROPUS ROBUSTUS
MOLAR SURFACE AREA / MM^2	About 100	About 200
ENAMEL	Thinner	Very thick
JAW MUSCLES	Smaller	Very large

a **DEDUCE** the diet of Paranthropus robustus, giving two reasons. [3]

b **EXPLAIN** why smaller teeth in Homo sapiens fit an omnivorous diet. [2]

c **SUGGEST** a limitation of inferring diet from teeth alone. [1]

d **STATE** one advantage of using digital models of fossils. [2]

Section D: Longer answer

6 MARKS

16 **EXPLAIN** how different modes of nutrition allow many species to live in one ecosystem. [6]

[B4.2.1, B4.2.3–6]